Student Career Advisor

AI mentorship platform using real developer survey statistics, interactive dashboards, career comparison, and learning roadmaps.

Project Type	Web-based AI career guidance platform
Data Source	Stack Overflow Developer Survey 2024
Records Processed	49,191 developer responses
Career Profiles	19 developer roles
Generated On	2026-05-24

1. Project Idea

The project is a Student Career Advisor website. Its purpose is to help students understand technology career paths using both conversational AI and real-world developer statistics. Instead of giving generic advice, the system grounds its recommendations in cleaned survey data from professional developers.

Core idea: a student can ask about a track such as Data Scientist, Back-End Developer, Cybersecurity Engineer, or AI / ML Engineer. The website detects the track, answers through an AI advisor, and shows visual data about salary, experience, skills, education, and employment.

2. What We Built

- A Flask backend that serves the website, exposes the survey JSON, and handles chat requests.

- A Gemini-powered AI advisor that provides friendly career guidance when an API key is configured.

- A role-matching system that detects career paths from user messages using aliases and canonical role names.

- An interactive dashboard that shows salary distribution, coding experience, popular languages, databases, education, and employment type.

- A career comparison modal for side-by-side role analysis.

- A Learning Roadmap Visualizer that builds an ordered skill tree from the survey-backed role profile.

3. Dataset and Preprocessing

The raw dataset is the Stack Overflow Developer Survey 2024. The raw CSV contains many columns, but the project focuses on fields that are useful for career guidance: developer type, languages, databases, frameworks, platforms, years of coding, education, employment, country, salary, age, learning resources, and industry.

- Selected only the career-relevant columns from the raw survey.

- Handled missing values by computing each metric from available responses only.

- Split multi-value survey fields such as languages and databases by semicolon.

- Cleaned years of coding so numeric experience statistics could be calculated.

- Capped unrealistic salary outliers before computing salary percentiles.

- Built one profile per supported developer role and saved the result in student_advisor_data.json.

4. System Architecture

Layer	Responsibility
Frontend	HTML, CSS, and JavaScript single-page interface with chat, dashboard, comparison modal, and roadmap visualizer.
Backend	Flask app in app.py serving static files, survey data, and /api/chat.

Knowledge Base	student_advisor_data.json containing role profiles, aliases, global stats, salaries, technologies, education, and employment data.
AI Layer	Google Gemini model receives grounded system instructions based on the matched role profile.
Visualization	Custom dashboard bars and Mermaid.js flowcharts for learning roadmaps.

5. Main User Flow

- The student opens the website and sees suggested career tracks.

- The student clicks a role chip or asks a question in the chat.

- The backend matches the message to a career role when possible.

- The AI advisor answers using exact statistics from the matched role profile.

- The dashboard updates with salary, experience, technology, education, and employment insights.

- The roadmap section appears with an ordered skill path and confidence sliders.

6. Supported Career Profiles

Role	Responses	Median Salary	Median Experience	Top Languages
Full-Stack Developer	12,351	$70,794	14.0 yrs	JavaScript, HTML/CSS, SQL
Back-End Developer	6,453	$78,390	15.0 yrs	SQL, JavaScript, Python
Software Architect	2,684	$100,700	23.0 yrs	JavaScript, SQL, HTML/CSS
Front-End Developer	1,974	$61,362	10.0 yrs	JavaScript, HTML/CSS, TypeScript
Desktop / Enterprise Developer	1,919	$72,815	20.0 yrs	SQL, JavaScript, HTML/CSS
Mobile Developer	1,391	$69,609	12.0 yrs	Kotlin, Swift, JavaScript
Embedded / IoT Developer	1,274	$78,926	15.0 yrs	C, Python, C++
Engineering Manager	1,068	$130,000	20.0 yrs	JavaScript, SQL, Python
DevOps Engineer	1,053	$87,011	15.0 yrs	Bash/Shell (all shells), Python, SQL
Data Engineer	770	$80,412	12.0 yrs	Python, SQL, Bash/Shell (all shells)

7. Learning Roadmap Visualizer

The newest feature adds a visual ordered skill tree for each selected track. The roadmap is generated from the same trusted survey profile used by the dashboard. It takes the top languages, database, frameworks, and platforms for that role, removes duplicates, and caps the list to a concise learning path.

- Each roadmap is rendered as a Mermaid.js flowchart.

- Students rate their confidence in every skill from 0% to 100%.

- Skills at 80% or above are treated as mastered.

- Skills between 40% and 79% are treated as currently learning.

- The first skill below 80% becomes the next priority.

- Progress is saved in localStorage per role, so it remains after refresh.

Example Roadmap

For Data Scientist: Python -> SQL -> PostgreSQL -> Flask -> FastAPI -> Docker

8. Global Technology Snapshot

Category	Top Items
Languages	JavaScript, HTML/CSS, SQL, Python, Bash/Shell (all shells), TypeScript
Databases	PostgreSQL, MySQL, SQLite, Microsoft SQL Server, Redis, MongoDB
Frameworks	Node.js, React, jQuery, Next.js, Express, ASP.NET Core

9. Project Files

File	Purpose
app.py	Main Flask server, role matching, Gemini chat integration, API routes.
static/index.html	Single-page app structure and dashboard sections.
static/script.js	Frontend behavior: chat, dashboard rendering, comparison modal, roadmap visualizer.
static/style.css	Application layout, dashboard design, responsive styling.
preprocess.py	Data cleaning and transformation script.
student_advisor_data.json	Preprocessed career knowledge base used by backend and frontend.
test_app.py	Basic route and role-matching verification script.

10. Testing and Verification

- Verified that the app serves the homepage route.

- Verified that the preprocessed JSON endpoint returns role profile data.

- Verified that chat messages can be matched to known developer roles.

- Checked JavaScript syntax after adding the roadmap feature.

- Validated roadmap generation against real roles such as Data Scientist, AI / ML Engineer, Back-End Developer, and UX / UI Designer.

11. Limitations and Future Work

- The AI chat requires a valid Gemini API key in the .env file.

- Roadmaps only use skills available in the current survey profile, so missing libraries are not invented.

- The salary data is global and should be interpreted as survey-based guidance, not a guaranteed salary.

- Future improvements could add user accounts, downloadable personal study plans, more charts, and country-specific salary filters.

Conclusion

This project combines real data, AI mentorship, and interactive visual learning tools to help students explore technology careers with more confidence. The result is not just a chatbot, but a complete career guidance dashboard that explains each path, compares roles, and helps students plan what to learn next.